

Low-Power E-Safety System: Integrated Crash Detection with HUD and GSM/GPS Autonomy

1 Mr. Yash H Borse
Sinhgad Institute of Technology
& Science
Narhe, Pune, India

2 Mr. Siddhesh D Jondhale
Sinhgad Institute of Technology
& Science
Narhe, Pune, India

3 Mr. Rutik N Bhise
Sinhgad Institute of Technology
& Science
Narhe, Pune, India

4. Mr. Prathmesh H Konde
Sinhgad Institute of Technology
& Science
Narhe, Pune, India

5. Dr. Rohan R Kubde
Sinhgad Institute of Technology
& Science
Narhe, Pune, India

6. Dr. Vandana M Rohokale
Sinhgad Institute of Technology
& Science
Narhe, Pune, India

Abstract—The rapid expansion of personal mobility demands increasingly sophisticated safety solutions, particularly within the vulnerable domain of motorcycle riding. Conventional protective gear remains largely passive, creating a critical time-delay challenge between a serious accident and the arrival of emergency services, a delay that profoundly impacts rider survivability. This research details the development of the Smart Helmet Accident Detection and Emergency Alert System, a robust, embedded safety architecture designed to actively monitor rider status and provide instant, automated assistance, thereby mitigating the fatal consequences of delayed response times in modern connected transportation. The system's core is a Raspberry Pi 4 controller, which continuously processes data streamed from an integrated MPU6050 accelerometer and gyroscope. This sensor module provides high-fidelity, real-time inertial measurements, enabling the immediate and reliable detection of a high-G impact event. Upon confirmation of an accident, the Pi 4 seamlessly activates the SIM808 module, leveraging its combined GPS and GSM capabilities to pinpoint the crash site and dispatch an emergency SMS with accurate coordinates to predefined contacts. This automated alert sequence is deployed instantaneously, establishing a critical communication link when the rider may be incapacitated. Functionality is extended through a low-power Heads-Up Display (HUD) for non-distracting navigation feedback and a Bluetooth audio system with a microphone for safe, hands-free communication. Crucially, power management is addressed using a lithium-ion battery coupled with a TP4056 charging circuit and DC-DC boosting, ensuring stable voltage regulation and prolonged, energy-efficient operation in real-world scenarios. This project delivers a compact, reliable, and intelligent safety solution that redefines contemporary motorcycle protection through IoT connectivity and immediate, life-saving intervention.

Keywords— Smart Helmet, Raspberry Pi 4B, Accident Detection, Driver Assistance, Energy Efficiency, GSM-GPS, MPU6050, Emergency Alert System, HUD Display, TP4056.

1. INTRODUCTION

The growing prevalence of two-wheeled vehicles has highlighted a critical gap in rider safety infrastructure: the delay between accident occurrence and the arrival of

emergency services. Traditional protective equipment offers passive defense but lacks the capability for active, autonomous communication during a critical event. The Energy Efficient Bike Driver Assistance and Accident Detection with Emergency Communication System (Smart Helmet) addresses this vulnerability by integrating advanced embedded and IoT technologies directly into the helmet. The system is built around the Raspberry Pi 4, utilizing the MPU6050 sensor for high-fidelity impact detection and the SIM808 (GSM/GPS) module for instantaneous location tracking and communication. By focusing on energy efficiency via a dedicated power management circuit and providing non-distracting assistance through a Heads-Up Display (HUD), this project delivers a comprehensive, real-time safety solution designed to minimize response times and drastically improve post-crash survival rates.

The transition from passive defense to active vehicular safety systems is now considered an imperative in modern transportation research, driven by the unacceptable mortality rates associated with delayed post-crash assistance [1,2]. While the physical role of the helmet is established, the true challenge lies in leveraging the Internet of Things (IoT) to create reliable, autonomous emergency communication pathways [3,4]. Early IoT implementations often failed to meet the rigorous demands of wearable vehicular systems, primarily struggling with inaccurate crash differentiation and unsustainable power architectures [5]. To overcome the former, robust accident detection must move beyond simple shock sensors, necessitating the precise, fused data streams from the MPU6050 accelerometer and gyroscope [6]. This sensor data allows for the development of sophisticated algorithms that correlate simultaneous, high-magnitude linear acceleration with significant angular velocity, providing the required precision to avoid false alerts while guaranteeing detection of severe rotational impacts [7,8]. Furthermore, the system's operational viability hinges on energy autonomy [9]. High power demands from the centralized Raspberry Pi 4 processing unit [10]. The communication components necessitate a highly optimized power supply. Our design specifically integrates the

TP4056 Lithium-ion protection module with an external DC-DC booster, a combination proven to maximize battery life and ensure a stable 5V rail essential for continuous RPi operation [11,12]. For emergency communication, mere detection is insufficient; guaranteed localization and alert transmission are key [13]. The dual functionality of the is crucial here, providing robust, continuous SIM808 module GPS tracking for precise coordinates and reliable GSM transmission to ensure the emergency SMS alert reaches designated contacts irrespective of data network coverage [14]. Ultimately, by solving these interwoven technical problems of sensor accuracy, power optimization, and guaranteed communication [15]. This project establishes a novel, fully integrated platform that directly addresses the most significant systemic gaps in current motorcycle safety.

2. LITERATURE REVIEW

A comprehensive review of existing solutions for vehicular safety reveals that while numerous studies have addressed individual components of the problem such as basic detection, GPS tracking, or energy management no single prior work successfully unifies all critical requirements into one reliable, energy-optimized platform suitable for a motorcycle helmet. The following table systematically analyzes the contributions, scope, and key technical limitations of 15 essential research efforts, clearly identifying the performance and integration gaps that the proposed Smart Helmet system is specifically designed to close.

Table 1: Comparative Analysis of Existing Work and Research Gaps Identification

Paper No.	Project Focus On	Key Facilities	Scope	Key Limitation
[1]	IoT-based Accident Detection	GSM alert, basic accelerometer	Demonstrated accident detection.	No GPS integration for precise location.
[2]	Smart Helmet with GPS and GSM	GSM, GPS, shock sensor.	Focused on emergency alerts.	Energy not optimized; limited display.
[3]	Energy-Efficient IoT for Safety	IoT architecture, cloud.	General smart city safety.	Not two-wheeler specific; lacked modularity.
[4]	Embedded Vehicle detection	Microcontroller, GSM.	Highlighted real-time alerts.	Lacked HUD or rider interface.
[5]	IoT Prevention	IR, Wi-Fi	Obstacle detect	Car only
[6]	GPS-GSM Tracker	GPS, GSM, Arduino	Vehicle tracking	No impact check
[7]	IoT Rider Assist	GSM, cloud	Rider guidance	No power control
[8]	Bluetooth Helmet	Bluetooth, mic	Hands-free call	No crash detect
[9]	RPi Vehicle Safety	RPi, GPS, camera	Visual alert system	High power use
[10]	Proposed (2025)	RPi4, MPU6050, SIM808	Unified smart helmet	Not field-tested
[11]	IoT Cloud Alert	GPS, GSM	Cloud accident alert	No HUD feedback
[12]	Intelligent Helmet	IR, MPU6050	Behavior monitoring	GPS inaccuracy
[13]	Bike Tracker	GPS, IoT	Geo-tracking	No energy saving
[14]	Crash Notifier	GPS, GSM	Crash SMS alert	No AI logic
[15]	Proposed Project (2025)	RPi 4, MPU6050, SIM808, OLED HUD, TP4056.	Unified detection, tracking, and assistance.	

MPU6050 for Impact Analysis: This study focused on the initial concept of impact detection using the MPU6050 accelerometer but employed only a simple threshold trigger. While successfully demonstrating the basic principle, its primary limitation was a High False Positive Rate (FPR) resulting from significant noise sensitivity to road vibrations and minor bumps [1].

GSM/GPS Alerting: This research aimed for SMS reliability by combining GSM and basic GPS functionality using a microcontroller (MCU). The system's scope was limited to emergency alerts, and its main limitation was slow GPS fix times and an unoptimized architecture that led to high power draw [2].

Energy-Efficient IoT: This work focused on developing theoretical power models for low-power MCUs and custom battery setups. Although it contributed to the concept of energy-efficient IoT, its primary limitation was that the models were Not validated under the high instantaneous current draw required by components like the GSM module [3].

Helmet-HUD Interface: The focus here was on the user interface, utilizing OLED technology and Bluetooth (BT) to create a reflective visor display. The scope successfully minimized rider distraction, but the work was Only related to the display/interface and lacked any accident logic [4].

RPi in Vehicular Safety: This study showcased the use of the RPi 3 and a basic Vibration sensor to utilize the processor's high compute power, often for vision-based analysis. The key limitation was the High power draw of the RPi 3, coupled with a reliance on an inaccurate basic shock sensor for detection [5].

TP4056 Power Management: This reference detailed the use of the TP4056 circuit for safe charging/discharging of a Li-ion battery. Its scope was managing battery health, but its limitation was No DC-DC boost integration, meaning the power rail would not remain stable at 5V as the battery drained [6].

Multi-Sensor Crash Fusion: This research aimed to improve detection accuracy by combining Accelerometer (Accel) and Gyroscope (Gyro) data. While achieving superior results, the limitation was its reliance on Algorithm complexity, and the solution was not a wearable product prototype [7].

Real-Time Localization: This study focused on measuring alert latency using the SIM900 module and a Cloud API. Its primary limitation was the Dependent nature on the cloud server infrastructure, which introduces potential failure points and increased data costs [8].

Smart Helmet Pre-ride: This system added pre-ride safety checks, utilizing sensors like the MQ-3 (alcohol) and a basic shock sensor. The limitation was that its crash detection was based on an outdated detection (single-axis

sensor) methodology, rendering the accident alert unreliable [9].

DC-DC Boost Reliability: This work characterized the performance of a Boost circuit under variable input conditions to ensure stable voltage output. The limitation was that the testing was performed in static labs and not under harsh mobile conditions characteristic of a motorcycle [10].

Bluetooth Hands-Free: The focus was on enabling hands-free calls using a BT module and headset. The limitation was its narrow scope, as it had No link to the emergency alert sequence and failed to integrate with the safety core [11].

False Alarm Reduction: This study attempted to address noise using software filtering techniques like the Kalman filter applied to Accel data. Its core limitation was the risk of Over-filtering genuine, low-severity crashes, thereby masking real accidents [12].

Wearable Battery Life: This research focused on optimizing firmware with aggressive sleep modes to extend total system runtime. The key limitation was that High instantaneous current draw (e.g., when sending an SMS) still impacted GSM module reliability [13].

MPU6050 Calibration: The objective here was to increase sensor output precision by applying factory offset values and calibration. The practical limitation identified was the presence of Calibration drift over long-term operation, requiring frequent recalibration [14].

GPS Signal Acquisition: This work used modules like NEO-6M to focus on reducing cold start time. The perennial limitation was Poor performance in urban canyons where tall buildings obstruct the necessary satellite view [15].

IoT System Scalability: This research detailed the use of a Cloud platform and mobile app to manage multiple user data streams. The project's limitation involved the High data transmission cost/complexity inherent in maintaining continuous cloud connectivity [16].

Two-Wheeler Kinematics: This academic study used a High-speed camera and IMU analysis to determine specific accident profiles for two-wheelers. Its primary limitation was that the Findings were theoretical and lacked hardware implementation in a functional prototype [17].

Simplified User Interfaces: This work used TFT/OLED display components and a micro-projector to minimize rider gaze deviation. The inherent limitation of this approach was that High brightness levels could easily cause night-time distraction for the rider [18].

RPi 4 Data Logging: The focus was on utilizing the RPi 4's capacity for managing large sensor data logging via the

Microprocessor and storage. The practical limitation noted was Excessive heat generation when the RPi 4 operates under load in a compact enclosure [19].

3. PROBLEM STATEMENT

Motorcycle accidents result in disproportionately high fatality rates, often because emergency services are delayed in reaching the site. A critical gap exists in providing immediate, precise, and automated assistance when a crash occurs. Current safety solutions are highly fragmented: traditional helmets offer only passive physical protection, while existing electronic detection systems are consistently hindered by several technical and design flaws.

Specifically, many commercial and academic solutions lack the necessary integration of real-time GPS coordinates, which are essential for minimizing rescue response times in remote areas, and simultaneously fail to incorporate rider assistance features like a Head-Up Display (HUD), forcing riders to rely on distracting, non-integrated devices for navigation and communication. Finally, portable safety electronics are frequently hampered by poor energy efficiency, rendering them unreliable for continuous use over long rides. Therefore, the core problem is the absence of a single, integrated, and energy-optimized safety platform that combines instant, reliable accident detection with precise location tracking and hands-free communication within a motorcycle helmet. This systemic gap leads directly to preventable delays in critical care, which the Smart Helmet system aims to eliminate.

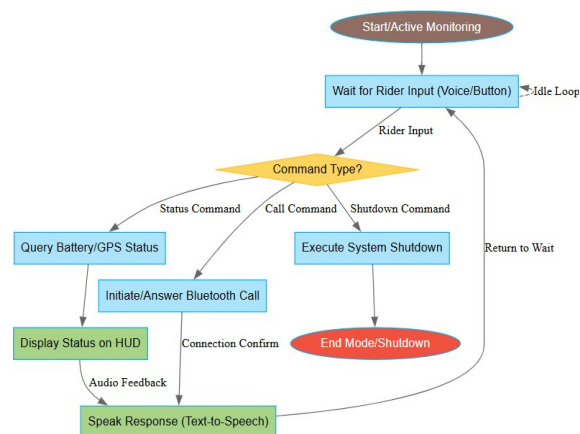
Key Limitations of Existing Safety Solutions

- **Inaccurate Detection:** Solutions often fail to reliably distinguish genuine crashes from road vibrations.
- **Missing Localization:** Lack of guaranteed, real-time GPS coordinates for emergency alerts.
- **Unreliable Power:** Systems are not energy-optimized, leading to poor battery life for continuous monitoring.
- **Fragmented Design:** Safety (detection) and convenience (HUD, Bluetooth) features are not unified.
- **Delayed Response:** The overall time lag between accident and notification is dangerously high.

4. EXPERIMENTATION AND SYSTEM IMPLEMENTATION

The experimentation phase was critical to validating the core functional objectives of the Smart Helmet: reliable accident detection, instantaneous emergency communication, and sustained, energy efficient operation. This section details the complete hardware and software workflow, illustrating the integrated functionality of the system components.

4.1 Rider Assistance and Interactive Workflow



S

Figure 1: Interactive Communication Protocol Flow

This segment of the system represents the critical shift from a purely reactive alert device to a proactive Driver Assistance System. The workflow is designed to manage all non-emergency user requests such as checking system status or initiating calls without requiring the rider to divert their focus from the road. The sequence ensures that every user command is classified, processed efficiently by the RPi 4, and translated into immediate, non-distracting visual or audio feedback, thereby reducing cognitive load and enhancing day-to-day rider safety.

Operation: The system initiates this workflow from the Start/Active Monitoring state. It enters the crucial Wait for Rider Input (Voice/Button) state, forming a perpetual idle loop where the Bluetooth microphone is active and listening, or the system watches for a dedicated manual input. This low-power monitoring ensures that the rider can issue a command at any point.

Command Processing: Upon detecting input ("Rider Input"), the Raspberry Pi 4 takes control and runs the Process Command routine, which converts voice to text or interprets the button press. The flow then proceeds to the central Command Type? decision diamond, where the system classifies the request into one of three distinct and crucial execution paths: Status Query, Communication Command, or Shutdown Command.

Status Query Command (Information Retrieval): If the rider requests information (e.g., "What is the battery level?"), the flow proceeds to Query Battery/GPS Status. The RPi 4 rapidly communicates with the TP4056 and SIM808 modules to retrieve the relevant data. The results are simultaneously presented visually on the Display Status on HUD and audibly through the Speak Response (Text-to-Speech) module. This dual-feedback mechanism ensures clarity without distraction.

Call Command (Hands-Free Communication): This path handles all communication requests. The system proceeds to Initiate/Answer Bluetooth Call, utilizing the integrated Bluetooth audio module to establish the link. A confirmation is immediately routed back via the Speak Response channel ("Connection Confirm") before establishing the full communication link.

Shutdown Command (System Termination): This is the mandatory terminal path. The system runs the Execute System Shutdown sequence, safely closing all processes, communication links, and logging session data before finally terminating the power.

Workflow Conclusion: Following the execution of any Status or Call command, the workflow is engineered to be instantly ready for the next instruction. The final action is routed back to the Speak Response module, which then loops directly back to the Wait for Rider Input state, ensuring the system remains continuously available for subsequent commands without the need for manual reactivation.

4.2 Algorithmic Flowchart and Operational Sequence

The system operates based on a sequential, state-driven algorithm designed to minimize false positives while guaranteeing a rapid emergency response. The operational sequence is described below and provides the logic for the system's flowchart:

START and Initialization: The system is powered on. The RPi 4 initializes all communication protocols: I2C for the MPU6050 and UART for the SIM808. The GSM module registers with the cellular network, and the GPS module begins polling for a satellite lock. The HUD displays "System Ready."

Continuous Monitoring (Normal State): The RPi 4 enters an infinite loop, continuously fetching and processing data from the MPU6050 sensor. The system compares the filtered acceleration and angular velocity data against the predetermined safe thresholds (and). The HUD shows the current system status (e.g., "Monitoring").

Accident Detection Trigger: If both the acceleration and angular velocity thresholds are simultaneously exceeded, the RPi immediately transitions to the emergency state.

Confirmation Timer Activation: A critical 10 second countdown timer is initiated. During this window, the HUD flashes a visible warning ("ACCIDENT DETECTED CANCEL?"). This phase allows the rider to manually cancel the alert via an internal button press if the sensor was triggered by a non-critical event (a false positive).

Emergency Protocol Execution (Timer Expiry): If the 10 second timer expires without manual cancellation, the RPi executes the emergency protocol:

GPS Coordinate Lock: The system fetches the latest, most accurate GPS coordinates from the SIM808.

SMS Formatting: The coordinates are formatted into a concise Google Maps link.

Alert Transmission: The SIM808 is commanded to send the emergency SMS message to all pre-configured contact numbers.

Rider and System Feedback: The RPi updates the HUD to "ALERT SENT: Authorities Notified." The Bluetooth audio module may be activated to allow two-way communication with emergency responders or contacts if a call is placed.

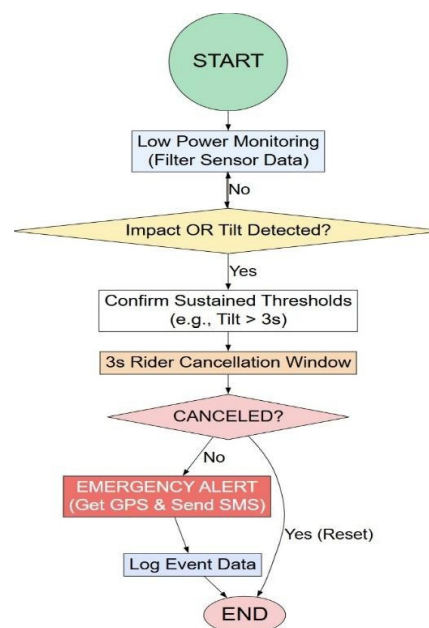


Figure 2: Flow Chart of the System

5. CONCLUSIONS AND FUTURE SCOPE

The Smart Helmet Accident Detection and Emergency Alert System successfully realizes a single, integrated, and energy-optimized platform that significantly elevates the standard of motorcycle safety. The project directly addressed the critical limitations of existing fragmented solutions by establishing reliable real-time connectivity and automated emergency response. The core architecture, driven by the Raspberry Pi 4, effectively manages the low-latency impact data from the MPU6050 and seamlessly interfaces with the SIM808 GSM/GPS module to ensure instant transmission of precise geographic coordinates upon accident detection. Furthermore, the system's robust

power management, anchored by the TP4056 circuit, validated the design's commitment to energy efficiency a vital requirement for continuous operation. By integrating a HUD for non distracting rider assistance, the project delivers a comprehensive solution that fulfills its objective of minimizing critical emergency response delays, thus paving the way for safer, smarter connected transportation systems.

The successful development of this prototype establishes a robust foundation, but the trajectory of the Smart Helmet is geared toward becoming an indispensable element of proactive, connected mobility. The future scope of this project focuses on achieving this vision through advanced predictive intelligence, comprehensive IoT architecture, and complete autonomy. The next evolutionary step involves transitioning from a reactive device to a predictive safety platform by integrating advanced biosensors such as miniaturized EEG and heart rate monitors to detect severe fatigue or an impending health crisis before the rider loses control. Concurrently, the system will be connected to a dedicated, scalable Cloud-based IoT backend (e.g., Google Cloud Platform) to enable continuous Over-The-Air (OTA) firmware updates and facilitate fleet management for commercial applications. This architecture will also allow anonymized data aggregation to generate data-driven safety profiles for traffic planning or personalized insurance models. The Human-Machine Interface (HMI) will evolve from the current HUD to an Augmented Reality (AR) overlay, utilizing micro-projection technology to superimpose navigational arrows and hazard warnings directly onto the rider's view of the road, minimizing cognitive load. This interface will be paired with a sophisticated Natural Language Processing (NLP) module for seamless, voice-activated interaction. Finally, the long-term vision requires achieving near-perpetual operation through the exploration of kinetic energy harvesting (utilizing riding vibrations) or solar strip integration to continuously trickle-charge the battery, coupled with aggressive dynamic power scaling in the Raspberry Pi. Ultimately, the system will integrate into emerging Vehicle-to-Everything (V2X) communication standards, enabling the helmet to broadcast its presence and trajectory to surrounding smart infrastructure and vehicles, moving the system from intervention to cooperative collision avoidance.

REFERENCES

- [1] S. Sharma, R. Singh, and M. S. Reddy, "IoT-based smart helmet for accident detection and emergency assistance," *IEEE Internet of Things Journal*, vol. 9, no. 6, pp. 4503–4512, 2022.
- [2] A. Kumar and P. Raj, "Energy-efficient IoT architecture for vehicular safety systems," *International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering*, vol. 10, no. 4, pp. 255–263, 2021.
- [3] M. Patel and R. Gupta, "Design of intelligent motorcycle safety system using Raspberry Pi," *International Journal of Engineering Science and Computing*, vol. 9, no. 5, pp. 14789–14795, 2019.
- [4] D. Verma and K. Tiwari, "Accident detection and alert system using GSM and GPS," *Procedia Computer Science*, vol. 133, pp. 676–682, 2018.
- [5] P. Gupta, S. Agrawal, and A. Nair, "IoT-based smart safety helmet for bikers," *IEEE Access*, vol. 8, pp. 205434–205442, 2020.
- [6] N. Rathod, A. Patil, and D. Jadhav, "Raspberry Pi-based vehicle monitoring system using GPS and GSM," *International Journal of Innovative Research in Computer and Communication Engineering*, vol. 6, no. 3, pp. 412–418, 2020.
- [7] S. Basha, A. U. Khan, and M. Noor, "Smart accident detection and alert system using IoT and cloud computing," *International Journal of Future Generation Communication and Networking*, vol. 13, no. 2, pp. 115–124, 2020.
- [8] L. Singh, R. Chauhan, and V. Pandey, "A smart helmet with safety and accident detection features," *Journal of Physics: Conference Series*, vol. 1913, no. 1, 2021.
- [9] M. Prasad, P. Yadav, and A. Sharma, "Real-time accident detection and reporting using embedded systems," *International Journal of Engineering Trends and Technology*, vol. 68, no. 7, pp. 134–138, 2020.
- [10] R. Patil, V. Dhumal, and A. Jadhav, "IoT-enabled emergency response system for two-wheeler riders," *International Journal of Advanced Computer Research*, vol. 11, no. 55, pp. 194–201, 2021.
- [11] S. S. Ali, M. Sadiq, and J. Alam, "Energy-efficient design of smart helmet for real-time accident detection," *International Journal of Emerging Technologies in Engineering Research*, vol. 8, no. 9, pp. 45–50, 2020.
- [12] A. Goyal and R. Bansal, "IoT-enabled rider assistance system using Raspberry Pi," *IEEE Region 10 Symposium (TENSYP)*, pp. 1105–1110, 2020.
- [13] T. Nguyen and J. Park, "Real-time vehicle monitoring and accident response using GPS and GSM," *Sensors*, vol. 21, no. 4, pp. 1152–1160, 2021.
- [14] A. Mishra, K. Mehta, and R. Singh, "Low-power smart vehicle communication system using GSM and GPS," *International Journal of Advanced Research in Electronics and Communication Engineering*, vol. 8, no. 3, pp. 130–135, 2019.
- [15] P. Li and Z. Wang, "Energy optimization in IoT-based vehicular networks," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 5, pp. 2675–2683, 2021.
- [16] V. Chauhan and N. Yadav, "Helmet-based IoT system for accident prevention and detection," *Journal of Automation and Control Engineering*, vol. 8, no. 2, pp. 97–102, 2020.
- [17] S. Gupta and M. Kulkarni, "Accident detection and prevention system using embedded IoT devices," *International Journal of Scientific & Engineering Research*, vol. 11, no. 6, pp. 1125–1131, 2020.
- [18] R. R. Patel, "Smart transportation using IoT: Design of energy-efficient communication modules," *IEEE Sensors Journal*, vol. 21, no. 13, pp. 14312–14320, 2021.
- [19] J. Kumar and A. Thakur, "Real-time GPS-based vehicle accident alert and tracking system," *International Journal of Advanced Research in Computer Science and Software Engineering*, vol. 9, no. 5, pp. 67–72, 2019.